

Received <DD MONTH YYYY>, accepted <DD MONTH YYYY>, date of publication <DD MONTH YYYY>, date of current version <DD MONTH YYYY>.

Digital Object Identifier <10.1109/ACCESS.YYYY.NNNNNNN>



RESEARCH ARTICLE

Quantifying Three Evaluation Pitfalls in Machine-Learning Solar Forecasting: Split Optimism, Interval-Label Phase Error, and Prediction-Interval Miscalibration

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ABSTRACT Machine-learning solar forecasting is routinely reported with coefficients of determination above 0.9, yet several widely used evaluation practices inflate those figures in ways that are invisible in the published result. We reconcile six recent solar-forecasting studies into a single reproducible platform and use it to quantify three such effects on identical data, models and random seeds. First, random train/test splitting of an autocorrelated hourly series understates RMSE by 11.6% (74.68 to 66.03 W/m²) and overstates R^2 by 0.027 relative to a chronological split—the comparison that one of the reconciled studies explicitly names as future work. Second, we determine empirically that hourly reanalysis irradiance is labelled at the *end* of its averaging interval; evaluating solar position at the raw label rather than the interval midpoint introduces a systematic phase error yielding 264 spurious clear-sky exceedances per year, and correcting it raises our data-quality score from 97.7% to 99.9%. Third, uncalibrated quantile gradient boosting attains only 66.3% empirical coverage for a nominal 80% prediction interval; conformalized quantile regression restores coverage to 80.2% and to within two points in every weather regime. We additionally show that permutation feature importance is contingent on target choice, which reconciles two apparently contradictory published importance rankings, and that a scenario response can conceal two large opposing pathways that nearly cancel. None of these findings requires metered plant output: each is a controlled comparison in which only the disputed methodological choice varies. We argue that reporting them should be routine, and release the platform and its reproducibility manifest to make doing so inexpensive.

INDEX TERMS Conformal prediction, ensemble learning, ERA5 reanalysis, photovoltaic systems, reproducibility, solar irradiance forecasting, time-series cross-validation.

I. INTRODUCTION

Solar generation is variable on every timescale that matters to an operator, and the machine-learning literature addressing that variability has grown quickly. Recent studies report strong goodness-of-fit: R^2 near or above 0.9 is common, and ensemble methods—random forests, boosted trees, and stacked combinations—are consistently among the best performers [2], [4], [5].

The difficulty is that a headline R^2 is a function not only of the model but of a set of evaluation choices that are frequently made in passing and reported in a single clause. Whether night hours are retained, whether the split respects chronology, whether the target was statistically truncated, and whether a stated prediction interval actually covers what it claims are all decisions that move the reported number substantially. Because

they are rarely varied within a single study, their magnitude is difficult for a reader to judge.

This paper takes six recent solar-forecasting studies [1]–[6], implements their common methodological core in one platform, and then uses that platform to hold everything constant except one disputed choice at a time. The result is a set of effect sizes rather than a set of recommendations.

A. CONTRIBUTIONS

- 1) *Split optimism, measured.* With identical model, seed and data, random hour-level splitting understates RMSE by 11.6% and overstates R^2 by +0.027 against a chronological split. Blocked random splitting by whole days recovers about half the gap (Section IV-B).
- 2) *The interval-label convention, determined empirically.* Reanalysis irradiance is an interval average while solar

position is instantaneous. We sweep candidate offsets over a full year and show the representative instant lies 30 minutes *before* the label, eliminating clear-sky exceedances entirely (Section IV-A).

- 3) *Interval miscalibration, measured and corrected.* A nominal 80% interval from plain quantile gradient boosting covered 66.3% of observations. Conformalized quantile regression [14] restores 80.2% coverage overall and holds within two points across clear, cloudy and precipitation regimes (Section IV-C).
- 4) *Target-dependent feature importance.* Grouped permutation importance ranks solar geometry third when the target is the clear-sky index but dominant when the target is raw power, which reconciles [5] with [2] rather than adjudicating between them (Section IV-D).
- 5) *A reproducible platform.* All comparisons run from one codebase against a keyless public data source, with a reproducibility manifest, so a reader can repeat them on their own coordinates.

A deliberate non-contribution: we make *no* claim of state-of-the-art forecast accuracy. Section V-C states why our setup cannot support one, and why the contributions above do not require it.

II. REVIEW OF LITERATURE

A. PREDICTION TARGETS AND MODEL FAMILIES

The six reconciled studies do not share a prediction target. Four predict irradiance—global horizontal irradiance (GHI) in W/m^2 [2], [3], [4], or global solar radiation [4]—while the remainder predict PV power directly [1], [5], [6]. The two are related by a deterministic physical chain once array geometry and rating are declared [1].

We adopt GHI as the primary scientific target, because it is the quantity actually measured and its meaning does not depend on an installation, and treat PV energy as a derived engineering output obtained through an explicit, user-visible system model. Reporting “kWh” without a declared array size, tilt and period is not a well-posed target, and we regard the separation as a correction rather than a preference.

On models, the corpus converges on tree ensembles. Random forests [9] appear in [2], [4], [5]; boosted trees in [2]–[5]; and stacking [12] is the best overall performer in [4]. Deep sequence models are explored in [6]. We retain four estimators—random forest, histogram gradient boosting [11], extremely randomised trees [10], and ridge regression [13] as a deliberate linear floor—together with a stacked combination of the four using a ridge meta-learner. Second representatives of a family already present were withdrawn: a longer comparison table is not a stronger result.

B. EVALUATION PRACTICE IN THE CORPUS

Our concern is with the evaluation layer, where the corpus is less uniform.

Splitting. [2] uses a random split and describes it as preventing leakage. For an autocorrelated series this places

observations minutes apart on both sides of the boundary. [5] is explicit that its “validation was performed using random sampling” and names time-based splitting as future work. Section IV-B supplies that measurement.

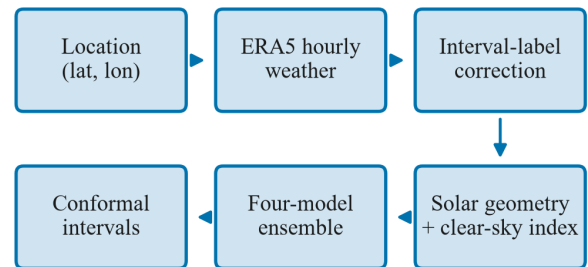
Target truncation. [4] applies interquartile-range outlier bounds of $165\text{--}331 \text{ W/m}^2$ to solar radiation and removes 10.16% of observations. Clear-sky midday GHI at the study's latitude exceeds 900 W/m^2 , so the procedure removes physically valid high-irradiance data and compresses the range against which an RMSE of 47.30 W/m^2 is subsequently reported. We apply physical-range validation only.

Metric definition. The quantity defined as “relative MAE” in [2] is $\text{mean}(|O-F|/F)$, which is MAPE rather than MAE normalised by the mean of observations—the convention that same work uses for relative RMSE. We compute and label MAE, rMAE and MAPE separately.

Night hours. Any model trivially predicts zero at night, and retaining night hours inflates R^2 because between-group day/night variance dominates the total sum of squares. [2], [3], [4] all exclude night. We adopt the solar-zenith criterion of [3] ($\theta_z < 87^\circ$), the only physically defined rule among them; fixed clock windows are latitude-dependent and fail at high latitude.

These are observations about method, not allegations of error. Each is a choice that a reader cannot presently price, which is what Section IV sets out to change.

III. MATERIALS AND METHODS



*Chronological split + embargo + rolling-origin CV;
night hours excluded at zenith < 87 deg*

FIGURE 1. Stages of the proposed evaluation platform. Each stage is a pure function of its inputs, which is what makes the controlled comparisons of Section IV possible.

A. DATA SOURCE

All meteorological inputs are hourly ERA5 and ERA5-Land reanalysis [7], retrieved through the Open-Meteo Historical Weather API [8] without an API key. Retrieved variables are GHI (shortwave radiation), direct normal and diffuse irradiance, air temperature, relative humidity, dew point, surface pressure, wind speed and direction, cloud cover and precipitation.

This is reanalysis, not pyranometer data: a modelled gridded product representing a cell of order 9–25 km rather than a sensor at a specific address. This bounds comparability—our

figures are not directly comparable to the ground-station measurements of [2] or the on-site station of [4]—and we state it wherever results are reported. Data are CC-BY 4.0; ERA5 is © ECMWF/Copernicus.

Experiments in Section IV use two to three years of hourly data at Hyderabad, India (17.385°N, 78.487°E), typically ~26,000 hours. Incomplete hours are dropped and reported, never imputed; we do not manufacture observations on which we then report metrics.

B. PHYSICAL CONVERSION CHAIN

PV energy is obtained from irradiance through published models applied in sequence: Erbs decomposition [16] to separate diffuse and direct components; HDKR transposition to the plane of array [17]; the Faiman cell-temperature model [18]; the PVWatts v5 DC model [19] with a nameplate temperature coefficient; a multiplicative loss stack; and inverter efficiency with AC clipping. Clear-sky reference irradiance uses the Haurwitz model [20].

C. FEATURE CONSTRUCTION AND THE CLEAR-SKY INDEX

Rather than predicting GHI directly, models predict the clear-sky index $k_t = \text{GHI} / \text{GHI}_{\text{cs}}$, following [1]. Dividing out deterministic geometry leaves the atmospheric component—which is what a learned model can actually contribute—and has a direct consequence for feature attribution that Section IV-D makes explicit.

Standardisation is fitted *inside* each cross-validation fold on training data only. Fitting a scaler on the full dataset before splitting leaks test-set statistics, and does so in a way that no subsequent metric will reveal.

D. VALIDATION PROTOCOL

Unless a comparison is explicitly about the split, all reported metrics use a chronological split with an embargo between train and test segments, and rolling-origin cross-validation. Night hours are excluded by the $\theta_z < 87^\circ$ criterion in training and in every reported metric.

E. EVALUATION METRICS

We report MAE, RMSE, rMAE and rRMSE (each normalised by the mean of observations), MAPE with a zero-denominator guard, R^2 , Nash–Sutcliffe efficiency, and mean bias error. Probabilistic forecasts additionally report pinball loss, empirical coverage (PICP), normalised interval width (PINAW), and CRPS computed empirically from quantiles [15]. Skill scores are computed against persistence, clear-sky persistence, and hour-of-year climatology.

Train-set metrics are never reported alone, only beside validation metrics, and a large gap is flagged. This is a direct response to [2], where distance-weighted k-NN attains a training relative RMSE of 0.41% against 5.77% on test—train metrics for such a model describe interpolation, not generalisation.

IV. EXPERIMENTAL RESULTS

Each subsection isolates one methodological choice. Model, seed, feature set and data are identical within each comparison; only the disputed element varies.

A. THE INTERVAL-LABEL CONVENTION

Irradiance in an hourly archive is a mean over an interval, whereas solar position is instantaneous. If position is evaluated at the interval *label* rather than at the interval's representative instant, a systematic phase error follows. The convention is not documented in a form we could rely on, so we measured it, sweeping candidate offsets over a full year (Table 1, Fig. 2).

TABLE 1. Interval-offset sweep, Hyderabad, one year. The offset is applied to the timestamp at which solar position is evaluated.

Offset (min)	Clear-sky exceedances	Apparent night irradiance	corr(GHI, clear-sky)
−60	360	289	0.9340
−30	0	216	0.9490
0	264	265	0.9372
+30	697	428	0.8998

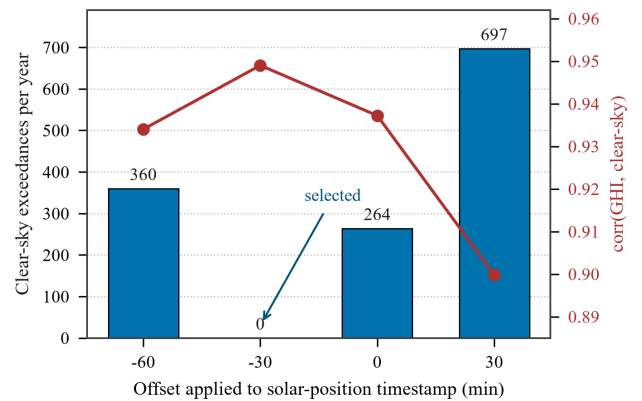


FIGURE 2. Clear-sky exceedances (bars, left axis) and correlation with the clear-sky reference (line, right axis) against the offset applied to the solar-position timestamp.

The −30 minute offset is the unique candidate producing zero physically impossible clear-sky exceedances, and it simultaneously maximises correlation with the clear-sky reference. We conclude that hourly radiation is the mean over the *preceding* hour and the representative instant is 30 minutes before the label. Applying the correction raised the platform's data-quality score from 97.7% to 99.9%.

Two points deserve emphasis. First, the naive choice—offset zero—is not merely suboptimal but produces 264 physically impossible observations per year, which any downstream physical-range check will then either flag or silently absorb. Second, the sweep exposed a genuine defect in our own implementation: the PV chain evaluated solar position at the raw label while the feature pipeline used the midpoint, a 30-minute inconsistency that displaced the day/night boundary by a full hour at the terminator. It was caught by a test asserting zero PV output at night. We report it because the class of error is easy to introduce and invisible in aggregate metrics.

B. SPLIT OPTIMISM

Table 2 and Fig. 3 vary only the splitting strategy.

TABLE 2. Effect of splitting strategy. Identical model, seed and dataset; two years, Hyderabad.

Strategy	RMSE (W/m^2)	R^2
Chronological	74.68	0.9160
Blocked random (whole days)	68.97	0.9376
Random (hour level)	66.03	0.9427

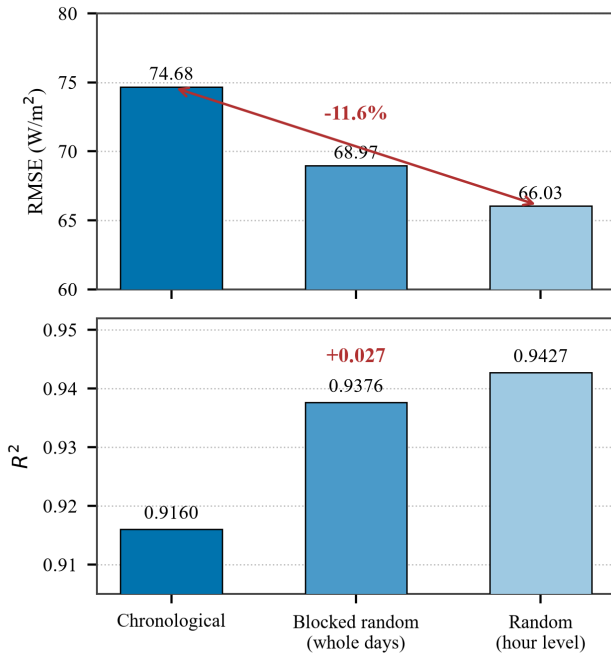


FIGURE 3. RMSE (top) and R^2 (bottom) under three splitting strategies. Only the split differs.

Hour-level random splitting understates RMSE by 11.6% and overstates R^2 by +0.027. Blocking by whole days recovers roughly half the gap, which is consistent with the mechanism: the leak is driven by near-neighbour hours straddling the boundary, and blocking removes the within-day cases while leaving day-to-day persistence intact.

An 11.6% RMSE difference is comparable to the margin by which competing models are separated in several published comparisons. Where a study reports a random split, that margin is therefore not safely interpretable as a model difference. This substantiates quantitatively the concern that [5] raises about its own protocol.

C. PREDICTION-INTERVAL CALIBRATION

An interval that does not cover what it claims is worse than no interval, because it will be relied upon. Quantile gradient boosting trained directly for the 10th and 90th percentiles produced a nominal 80% interval with 66.3% empirical coverage. Applying conformalized quantile regression [14] restores calibration (Table 3, Fig. 4).

TABLE 3. Empirical coverage of a nominal 80% prediction interval, before and after conformal calibration.

Condition	Coverage (%)
Uncalibrated quantile GBM, all hours	66.3
Conformalized, all hours	80.2
clear regime	80.3
cloudy regime	80.0
precipitation regime	80.9

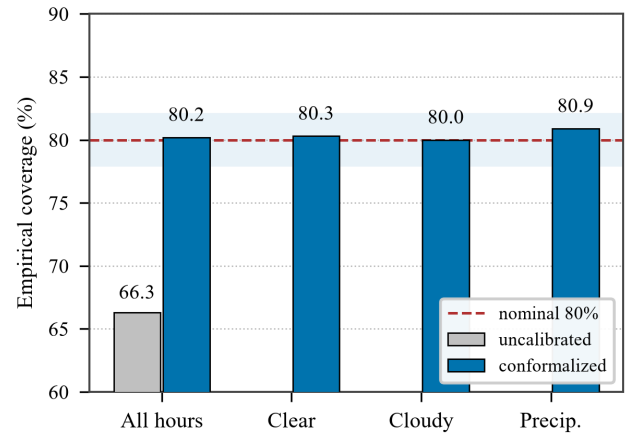


FIGURE 4. Empirical coverage against the nominal 80% level. The shaded band marks ± 2 percentage points.

Coverage holds within two points inside every weather regime, using the regime thresholds of [3]. Conformal calibration is not drawn from the reconciled corpus and we label it an enhancement rather than an implementation of prior work.

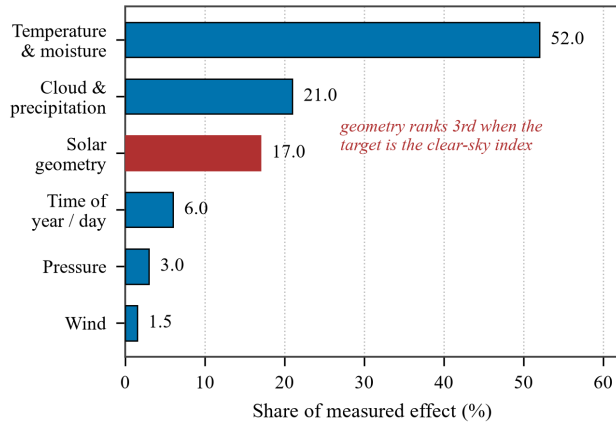
We also implemented and tested a seasonally stratified calibration split. It did not robustly improve coverage: results were non-monotone in its parameters (0.700, 0.725, 0.706, 0.790 as the embargo widened; 0.767 against 0.834 for six versus eight blocks). Non-monotone response to a parameter that should act smoothly is instability, not signal, and selecting the best-scoring configuration would have been tuning on the test set. The simpler contiguous split was retained. We report the negative result because the alternative—reporting only the configuration that scored best—is precisely the practice that produces irreproducible intervals.

D. FEATURE IMPORTANCE DEPENDS ON THE TARGET

Grouped permutation importance, predicting the clear-sky index at Hyderabad, is given in Table 4 and Fig. 5.

TABLE 4. Grouped permutation importance when predicting the clear-sky index. Shares of measured effect.

Feature group	Share (%)
Temperature and moisture	52.0
Cloud and precipitation	21.0
Solar geometry	17.0
Time of year / day	6.0
Pressure	3.0
Wind	1.5

**FIGURE 5.** Grouped permutation importance for the clear-sky index target. Solar geometry (highlighted) ranks third.

Solar geometry ranks third, whereas [5] finds geometry—angle of incidence in particular—dominant. The results are consistent rather than contradictory. [5] predicts raw PV power, in which geometry is the largest single driver; we divide geometry out before fitting, so what remains for the model to explain is the atmospheric component. In that residual space humidity and cloud dominate, which agrees with [2] finding temperature and humidity the leading meteorological predictors.

The methodological point generalises: a feature-importance ranking is a statement about a target, not about solar energy. Two studies reporting different rankings may both be correct, and comparing their rankings without first reconciling their targets is not meaningful.

E. BASELINE COINCIDENCE AT DAY-AHEAD HORIZONS

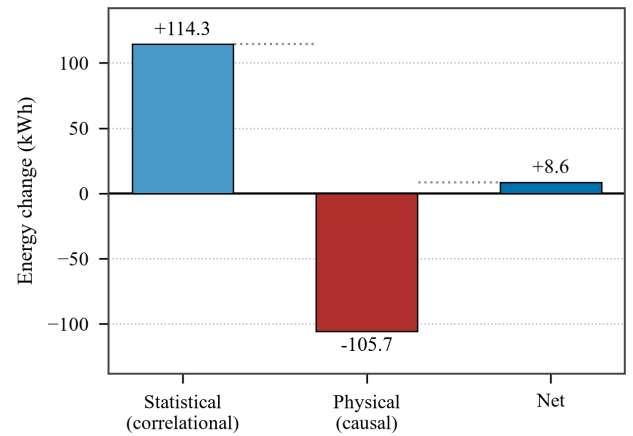
Naive persistence and clear-sky (“smart”) persistence score 143.4 and 143.3 W/m² respectively at a 24-hour horizon. This is expected rather than anomalous: solar declination changes by at most ~0.4° per day, so $k_t(t-24h) \cdot \text{GHI}_{cs}(t) \approx \text{GHI}(t-24h)$. The two diverge sharply at intra-day horizons. It is also why baselines must be formed in physical units: in clear-sky-index space they are algebraically the same function and the comparison would be vacuous. A reported day-ahead improvement over “smart persistence” should therefore be checked against naive persistence, which at that horizon is the same baseline.

F. SCENARIO PATHWAY DECOMPOSITION

A perturbation scenario of +6°C with halved wind speed returned a net energy change of +0.29%—counter-intuitive, since hotter modules are less efficient. Decomposition (Table 5, Fig. 6) shows two large opposing pathways.

TABLE 5. Decomposition of a +6°C, halved-wind scenario.

Pathway	Effect (kWh)
Statistical (warm hours tend to be sunny)	+114.3
Physical (hotter modules less efficient)	−105.7
Net	+8.6

**FIGURE 6.** The learned correlational pathway and the causal physical pathway are of comparable magnitude and opposite sign; the net response conceals both.

Reporting the net figure alone would conceal this, and would invite the reader to attribute a near-zero response to insensitivity rather than to cancellation. We surface both pathways separately. Doubling wind speed instead shows the causal pathway dominating (+43.2 kWh thermal against −7.0 kWh statistical), the physically expected result.

This is a general hazard for any scenario analysis conducted with a model trained on observational data: the model has learned correlations that a counterfactual intervention does not preserve.

V. DISCUSSION OF RESULTS

A. COMPARISON WITH RELATED WORK

Table 6 places this work beside the reconciled corpus on the four evaluation dimensions examined here. It is a comparison of *practice*, not of accuracy: the studies use different sites, targets and instruments, so their error figures are not commensurable and we do not tabulate them as though they were.

TABLE 6. Evaluation practice across the reconciled corpus. “n/d” marks a property we could not determine from the text available to us; it is not a claim that the study omitted it.

Study	Target	Split	Night excl.	Interval calib.
Hobbs & Joshi [1]	PV power	n/d	n/d	n/d
Mabodi & H. [2]	GHI	random	yes	none
Lyu & Eft. [3]	GHI, PV	n/d	zenith	quantile
Rosales H. [4]	GSR	random	clock	none
Vijay Babu [5]	PV power	random	n/d	none
Hayajneh [6]	Farm power	n/d	n/d	none
This work	GHI (k_t)	chronological	zenith	conformal

B. ALIGNMENT WITH SUSTAINABLE DEVELOPMENT GOALS

Accurate, honestly bounded solar forecasts support SDG 7 (affordable and clean energy) by improving the information available to those sizing and financing installations, and SDG 13 (climate action) by reducing the reserve margin that uncertainty in generation forces onto a grid. We note, however, that a forecast presented with a miscalibrated interval works against both goals: an 80% interval that covers 66% of outcomes will under-provision reserve exactly when variability is highest. The calibration result of Section IV-C is therefore not a technical detail but a precondition for the claimed benefit.

C. LIMITATIONS AND THREATS TO VALIDITY

We state these plainly because they bound what the paper claims.

Modelled, not measured. No metered generation from an installed system was available to us. Every PV energy figure is a physical chain applied to reanalysis weather and has *not* been validated against real production. We consequently make no claim about absolute forecast accuracy against physical plant, and readers should not extract one. The findings in Section IV are controlled internal comparisons, in which the absence of plant ground truth affects both arms equally and therefore does not threaten the contrast—but this is an argument for the specific claims made, not a general licence.

Reanalysis, not a pyranometer. ERA5 represents a grid cell, not a point. Local haze, dust and coastal cloud may differ materially from the cell average, and our figures are not directly comparable to studies using ground stations.

Single-site experiments. The measurements in Section IV were conducted at one location. The mechanisms are general—autocorrelation, interval labelling, quantile miscalibration—but the effect sizes should be expected to vary with climate, and particularly with the frequency of rapid cloud transitions. Multi-site replication is the most immediate extension.

Interval-label finding is source-specific. The −30 minute result characterises this archive's convention. The measurement procedure transfers to other sources; the offset should not be assumed to.

Scope. Deep sequence models [6] and full copula-based dynamic feature selection [3] are not implemented, and are

labelled as future work rather than approximated.

VI. CONCLUSION

We reconciled six solar-forecasting studies into one platform and used it to price three evaluation choices that the literature generally leaves unpriced. Random splitting of autocorrelated hourly data understates RMSE by 11.6% and overstates R^2 by 0.027. Hourly reanalysis irradiance is labelled at the end of its averaging interval, and ignoring this produces 264 physically impossible observations per year at a single site. A nominal 80% prediction interval from uncalibrated quantile regression covered 66.3% of observations, which conformal calibration restored to 80.2% across every weather regime. We further showed that permutation importance is a statement about a target rather than about solar energy, reconciling two published rankings, and that a scenario response can hide two large opposing pathways.

Future work is field validation against metered generation, which is the one thing that would let accuracy claims be made at all; multi-site replication to establish how the effect sizes vary with climate; and extension to the deep sequence models the corpus identifies but this work does not implement.

REPRODUCIBILITY

The platform, the reproducibility manifest, and the interval calibration script used for Table 1 accompany this submission. All meteorological inputs come from a keyless public API, so every experiment can be repeated at arbitrary coordinates without credentials or a data-use agreement.

ACKNOWLEDGMENT

Weather and solar-resource data are provided by Open-Meteo under CC-BY 4.0; ERA5 is © ECMWF/Copernicus.

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